MAST20030 Differential Equations
— Problem Sheet 6: Fourier Transforms —

Many problems involving Fourier transforms (including ones in this problem sheet) do not have closed form analytic solutions. In such cases, you may leave your final answer as an integral.

1. Consider the function
$$f(t) = \begin{cases} 0, & t < 0 \\ \pi/2, & t = 0 \\ \pi \mathrm{e}^{-t}, & t > 0 \end{cases}$$
   (a) Sketch $f(t)$ for all real values of t.
   (b) Find the Fourier transform of f.

2. Consider the function
$$f(t) = \begin{cases} 1, & 0 \le t \le 1 \\ 2, & 1 < t \le 4 \\ 0, & t > 4 \end{cases}$$
   (a) Sketch $f(t)$.
   (b) Find the Fourier transform of the even extension of $f(t)$.
   (c) Find the Fourier transform of the odd extension of $f(t)$.

3. Consider the function
$$f(t) = \begin{cases} \frac{\pi}{2} \cos t, & 0 \le t < \pi/2 \\ 0, & t \ge \pi/2 \end{cases}$$
   (a) Sketch $f_e(t)$, the even extension of f, for all real values of t.
   (b) Find the Fourier transform of f_e.

4. Consider the function
$$f(t) = -\mathrm{e}^{-2t} \qquad \text{for } t > 0$$
   (a) Sketch $f_o(t)$, the odd extension of f, for all real values of t.
   (b) Find the Fourier transform of f_o.

5. Consider the inhomogeneous differential equation

$$y'' + y = f$$

where

$$f(t) = \begin{cases} \sin t, & |t| \le \pi \\ 0, & |t| > \pi \end{cases}$$

(a) Find the Fourier transform of f.

(b) Using part (a), find the solution to the differential equation subject to the condition that the Fourier transform of y exists.

6. Consider an infinitely long beam on an elastic foundation with deflection u satisfying the nonhomogeneous differential equation

$$EI\frac{d^4u}{dx^4} + Ku = y$$

where the constant K is the "foundation modulus" (that is, the spring stiffness per unit x length) and EI the "flexural rigidity" of the beam and y is a prescribed loading (force per unit x length) given by

$$y(x) = \begin{cases} \omega_0, & |x| < 1 \\ \omega_0/2, & |x| = 1 \\ 0, & |x| > 1 \end{cases}$$

(a) Find the Fourier transform of y.

(b) Using part (a), find the solution to the differential equation subject to the condition that the Fourier transform of u exists.

7. Compute the Fourier transforms of the truncated waves

$$c(x) = \begin{cases} \cos(\alpha x) & -b \le x \le b, \\ 0 & \text{otherwise}, \end{cases} \qquad s(x) = \begin{cases} \sin(\alpha x) & -b \le x \le b, \\ 0 & \text{otherwise} \end{cases}$$

directly from the definition. Now compute them again using convolution and Heaviside functions. Are your answers consistent with the rules relating Fourier transforms and derivatives?

8. The definitions we have given for convolution in the Fourier and Laplace transform sections *do not agree*. The reason is that Laplace transforms are only defined on functions whose domain is $[0, \infty)$. Given any function $f(t)$, defined for all $t \in \mathbb{R}$, we can replace it by $\widetilde{f}(t) = f(t)H(t)$, so that its values for $t < 0$ are all 0. Show that the convolution of $\widetilde{f}(t)$ and $\widetilde{g}(t)$, using the definition given in the Fourier section, reproduces the convolution of f and g, using the definition given in the Laplace section.

9. Solve the following IBVPs for $x \in \mathbb{R}$ and $t \ge 0$, subject to the "boundary conditions" that the Fourier transform of the solution $u(x,t)$ exists, i.e., $\int_{-\infty}^{\infty} |u(x,t)|\,\mathrm{d}x < \infty$ for all $t \ge 0$:

(a) $\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}, u(x,0) = \mathrm{e}^{-x^2}$

(b) $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, u(x,0) = x\mathrm{e}^{-|x|/2}, \partial_t u(x,0) = 0$

(c) $\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}, u(x,0) = 20x\mathrm{e}^{-x^2}$

(d) $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, u(x,0) = 2\mathrm{sech}^2 x, \partial_t u(x,0) = 0$

(e) $\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}, u(x,0) = \frac{1}{1+x^2}$

(f) $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}, u(x,0) = \frac{x}{x^2+16}, \partial_t u(x,0) = 0$

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1. (b) $\widehat{f}(\omega) = \dfrac{\pi}{1+i\omega}$

2. (b) $\widehat{f}(\omega) = \dfrac{4\sin 4\omega - 2\sin\omega}{\omega}$

   (c) $\widehat{f}(\omega) = \dfrac{2i(2\cos 4\omega - \cos\omega - 1)}{\omega}$

3. (b) $\widehat{f}(\omega) = \dfrac{\pi\cos\frac{\omega\pi}{2}}{1-\omega^2}$

4. (b) $\widehat{f}(\omega) = \dfrac{2i\omega}{4+\omega^2}$

5. (a) $\widehat{f}(\omega) = \dfrac{2i\sin\pi\omega}{\omega^2-1}$

   (b) $y(t) = -\dfrac{i}{\pi}\displaystyle\int_{-\infty}^{\infty} \frac{\sin\omega\pi}{(1-\omega^2)^2} e^{i\omega t} d\omega$. This integral can actually be evaluated (by using Fourier methods outside the scope of this course) to give $y(t) = \begin{cases} \frac{1}{2}\pi\cos t, & t < -\pi \\ \frac{1}{2}(\sin t - t\cos t), & -\pi \le t < \pi \\ -\frac{1}{2}\pi\cos t, & \pi \le t \end{cases}$

6. (a) $\widehat{y}(k) = \dfrac{2\omega_0\sin k}{k}$

   (b) $u(x) = \dfrac{\omega_0}{\pi}\displaystyle\int_{-\infty}^{\infty} \frac{\sin k}{(EIk^4+K)k} e^{ikx} dk$

7. $\widehat{c}(k) = \dfrac{\sin b(k-\alpha)}{k-\alpha} + \dfrac{\sin b(k+\alpha)}{k+\alpha}$ and $\widehat{s}(k) = \dfrac{i\sin b(k-\alpha)}{\alpha-k} + \dfrac{i\sin b(k+\alpha)}{k+\alpha}$

8. Solutions to proof not provided.

9. (a) $u(x,t) = \dfrac{1}{\sqrt{4\pi\alpha t}}\displaystyle\int_{-\infty}^{\infty} e^{-\frac{(x-x')^2}{4\alpha t}} e^{-x'^2} dx'$

   (b) $u(x,t) = \dfrac{1}{2}\left((x-ct)e^{-\frac{|x-ct|}{2}} + (x+ct)e^{-\frac{|x+ct|}{2}}\right)$

   (c) $u(x,t) = \dfrac{20}{\sqrt{4\pi\alpha t}}\displaystyle\int_{-\infty}^{\infty} e^{-\frac{(x-x')^2}{4\alpha t}} x' e^{-x'^2} dx'$

   (d) $u(x,t) = \operatorname{sech}^2(x-ct) + \operatorname{sech}^2(x+ct)$

   (e) $u(x,t) = \dfrac{1}{\sqrt{4\pi\alpha t}}\displaystyle\int_{-\infty}^{\infty} e^{-\frac{(x-x')^2}{4\alpha t}} \frac{1}{1+x'^2} dx'$

   (f) $u(x,t) = \dfrac{1}{2}\left(\dfrac{x-ct}{(x-ct)^2+16} + \dfrac{x+ct}{(x+ct)^2+16}\right)$